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Model Deployment

By Prince Lesapo



What Is Model Deployment?

"Model deployment is the process of making a trained machine learning model available in a real-world environment so that it can make predictions or decisions on new data."

Sculley et al. (2015). Hidden Technical Debt in Machine Learning Systems. NeurIPS.

In plain terms: you built the model. Now you let the world use it.

Without deployment

The model lives in a notebook on your laptop.
Only you can run it.
It has no real-world impact.

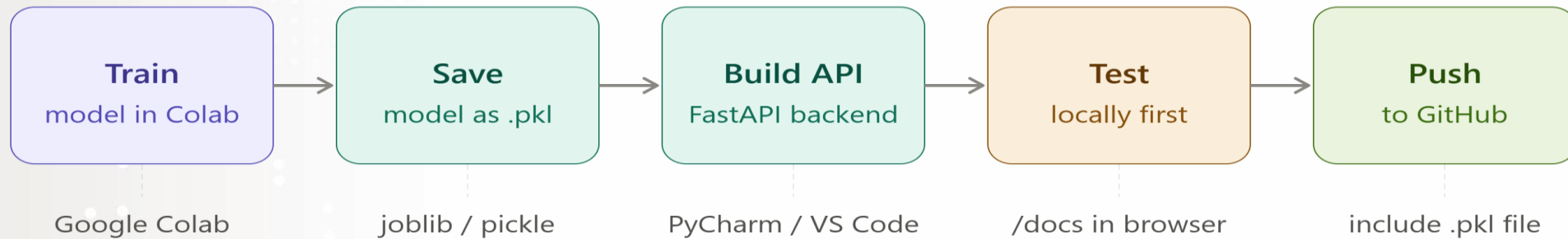
With deployment

The model lives on a server.
Anyone can send data and receive a prediction.
It runs 24/7 without you.



The Deployment Journey

From a trained model on your laptop to a live prediction anyone can access.





Step 1: Save Your Model

After training, your model exists only in memory. Saving it creates a file you can load anywhere, your laptop, a server, a colleague's machine.

What a saved model file is

A snapshot of everything the model learned during training.
Stored as a binary file, commonly .pkl or .joblib.
Load it once and make unlimited predictions without retraining.

Common tools: joblib and pickle

joblib is the standard for scikit-learn models.
Faster than pickle for large arrays and numpy objects.
One line to save. One line to load. That is all.

The most common deployment mistake

Pushing your code to GitHub without including the model file. The server receives the API code but has nothing to load. Every prediction call fails. The model file must be committed to your repository alongside your code.

Step 2: Set Up Your Environment

What is a virtual environment?

Think of it as a clean, fresh Python installation just for this project.

You install only what this project needs.

If another project needs a different version of a library, there is no conflict.

Industry standard practice, always use one.

PyCharm makes this automatic

Create a new FastAPI project in PyCharm and it sets up the virtual environment for you.

No manual commands needed.

VS Code requires you to create one manually, still straightforward but one extra step:

- `python -m venv venv`
- `venv\Scripts\activate`

Libraries you will need for this session:

pandas / numpy

Data handling

scikit-learn

The model

fastapi + uvicorn

The API framework

joblib

Load the saved model

shap

Explain predictions

Step 3: Build an API

API = Application Programming Interface

A way for two systems to communicate.

One side sends a request with data.

The other side processes it and sends a response back.

The sender never sees the logic behind the response.

Everyday example

When you check your bank balance on your phone:

Your app sends a request to the bank's API.

The bank's server processes it and sends your balance.

You never touched the bank's database directly.

Our ML API does the same thing. Someone sends applicant data. The server runs it through the model and sends back a prediction.

Why FastAPI?

The most popular Python framework for ML APIs. Automatically generates a testing interface at /docs so you can test predictions in the browser without writing frontend code. Fast to build, easy to test, used in production at major companies.



Step 4: Test Before You Push

Never push untested code. Test the API completely on your machine first.

What to test locally

Does the API start without errors?

Does it load the model file correctly?

Does a test prediction return the right format?

Does it handle wrong input gracefully?

FastAPI's /docs page lets you do all of this in a browser.

Why this step matters

If it does not work locally, it will not work on a server.

Debugging on a remote server is much harder than debugging locally.

A prediction endpoint that crashes silently is worse than one that never launched.

Test locally, fix locally, then push.

The golden rule: if you cannot show it working on your machine, you are not ready to deploy.



Step 5: Push to GitHub

What GitHub is

A platform for storing and sharing code.
Every deployment platform reads from GitHub.
Version controlled — every change is tracked.
Your project becomes accessible anywhere in the world.

What your repository must contain

Your API code file
Your saved model file (.pkl)
A requirements.txt listing all your libraries
A .gitignore to exclude files that should not be pushed

Check your .gitignore before pushing

Make sure the model file is not listed in .gitignore. If it is, GitHub will silently ignore it. Your deployment will launch, your API will start, and every prediction will fail.



Going Further: Explainable AI

A model that cannot explain itself cannot be trusted at scale.

Why explainability matters

In financial services, many countries legally require lenders to explain credit decisions.

You cannot tell someone they are denied without saying why.

This is called the right to explanation, and it applies to AI systems.

SHAP: SHapley Additive Explanations

A framework that tells you exactly which features drove a prediction and by how much.

Not just approved or denied, but WHY, with a value for each feature.

Based on Shapley values from game theory. Lundberg & Lee (2017).

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